

Model Lab

Every prior, same slate — 2026-09-17

SERVED PRIOR

Core (KNN+SVR)

PRIORS COMPARED

10

HELD-OUT PA

35,734

Held-Out Scores

Fitted on plate appearances before 2026-08-12, scored on the 35,734 that came after (actual HR rate 2.99%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5709	1.38×	0.02897	0.13366	0.00358
KNN comps Prior is the pooled rate of the nearest comparables.	0.5871	1.53×	0.02896	0.13333	0.00489
SVR Support-vector regression on the same feature matrix.	0.5765	1.42×	0.02904	0.13504	0.00821
Random forest Forest regression; a learned similarity metric.	0.5792	1.45×	0.02901	0.13440	0.00719
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5867	1.52×	0.02897	0.13337	0.00537
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5865	1.52×	0.02897	0.13338	0.00498
Linear regression Ordinary least squares on the same features.	0.5842	1.44×	0.02899	0.13405	0.00618
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5865	1.52×	0.02900	0.13373	0.00698
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5861	1.49×	0.02899	0.13367	0.00676
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5766	1.46×	0.02905	0.13483	0.00832

Blend weights (forest variant): KNN 0.90, SVR 0.00, RF 0.10

Shrinkage K per prior: league 247, knn 247, svr 247, rf 247, core 247, ensemble 247, linear 247, logistic 247, nn 247, xgb 247 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,819 PA, actual HR rate 2.54%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5097	1.03×	0.02478	0.11876	0.00632
KNN comps Prior is the pooled rate of the nearest comparables.	0.5639	1.13×	0.02484	0.12022	0.01300
SVR Support-vector regression on the same feature matrix.	0.5373	1.13×	0.02526	0.13130	0.02388
Random forest Forest regression; a learned similarity metric.	0.5495	1.13×	0.02509	0.12684	0.01942
Core (KNN+SVR) (served) Blend of KNN and SVR, no forest.	0.5629	1.24×	0.02485	0.12050	0.01305
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5630	1.34×	0.02486	0.12062	0.01316
Linear regression Ordinary least squares on the same features.	0.5640	1.34×	0.02501	0.12603	0.01708
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5654	1.24×	0.02495	0.12219	0.01587
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5709	1.44×	0.02491	0.12179	0.01553
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5437	1.03×	0.02537	0.12963	0.02451

Slate Agreement

Flat league · 3/6 shared 0/6 hit

Junior Caminero

Francisco Lindor

Juan Soto

Oneil Cruz

Elly De La Cruz

Héctor Rodríguez

SVR · 5/6 shared 0/6 hit

Junior Caminero

Juan Soto

Max Muncy

Kyle Schwarber

Oneil Cruz

Francisco Lindor

Core (KNN+SVR) · 6/6 shared 0/6 hit

Junior Caminero

Juan Soto

Max Muncy

Oneil Cruz

Jackson Merrill

Kyle Schwarber

Linear regression · 4/6 shared 0/6 hit

Emmanuel Rodriguez

Junior Caminero

Max Muncy

Juan Soto

Kyle Schwarber

Ben Ross

Neural net (torch) · 5/6 shared 0/6 hit

KNN comps · 6/6 shared 0/6 hit

18.7% Junior Caminero

16.5% Juan Soto

15.1% Max Muncy

15.0% Oneil Cruz

14.9% Jackson Merrill

14.9% Kyle Schwarber

Random forest · 5/6 shared 0/6 hit

22.5% Junior Caminero

19.7% Juan Soto

19.1% Max Muncy

18.4% Emmanuel Rodriguez

18.2% Kyle Schwarber

17.5% Jackson Merrill

Forest blend · 6/6 shared 0/6 hit

19.9% Junior Caminero

18.7% Juan Soto

18.3% Max Muncy

17.7% Oneil Cruz

17.3% Jackson Merrill

17.2% Kyle Schwarber

Logistic regression · 4/6 shared 0/6 hit

20.2% Emmanuel Rodriguez

20.0% Ben Ross

18.4% Junior Caminero

18.4% Max Muncy

17.8% Kyle Schwarber

17.4% Juan Soto

19.7%

18.6%

18.3%

17.7%

17.3%

17.1%

19.9%

19.0%

18.3%

18.0%

17.9%

17.5%

19.7%

18.7%

18.3%

17.6%

17.3%

17.2%

27.3%

22.4%

20.7%

20.0%

19.9%

18.8%

Jackson Merrill

Oneil Cruz

18.0%

Hitter by Hitter

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
1. Junior Caminero slate Tampa Bay Rays · vs RHP · 41 HR / 659 PA	18.7%	19.7%	22.5%	19.9%	19.9%	19.7%	20.0%	20.7%	20.9%	20.8%
2. Juan Soto slate New York Mets · vs RHP · 25 HR / 435 PA	15.1%	18.6%	19.7%	19.0%	18.7%	18.7%	18.4%	18.8%	18.8%	18.7%
3. Max Muncy slate Los Angeles Dodgers · vs LHP · 28 HR / 527 PA	14.8%	18.3%	19.1%	18.3%	18.3%	18.3%	18.4%	20.0%	18.9%	18.8%
4. Oneil Cruz slate Pittsburgh Pirates · vs RHP · 19 HR / 388 PA	15.0%	17.7%	18.2%	17.1%	17.7%	17.6%	17.2%	17.1%	17.7%	17.6%
5. Jackson Merrill slate San Diego Padres · vs RHP · 25 HR / 628 PA	14.8%	17.3%	16.8%	17.5%	17.3%	17.3%	17.3%	18.2%	17.7%	17.6%
6. Kyle Schwarber Philadelphia Phillies · vs RHP · 45 HR / 655 PA	14.8%	17.1%	18.4%	17.9%	17.2%	17.2%	17.8%	19.9%	17.9%	17.8%
7. Teoscar Hernández Los Angeles Dodgers · vs LHP · 15 HR / 450 PA	14.5%	17.0%	16.7%	17.1%	17.0%	17.0%	16.4%	16.5%	16.6%	16.5%
8. Shane Bieber slate Boston Red Sox · vs RHP · 25 HR / 450 PA	14.5%	17.0%	16.7%	17.1%	17.0%	17.0%	16.4%	16.5%	16.6%	16.5%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	XGBoost
8. Francisco Lindor New York Mets · vs RHP · 20 HR / 421 PA	16.5%	16.7%	17.5%	17.0%	16.8%	16.8%	17.0%	17.1%	16.9%	17.2%
9. Emmanuel Rodriguez Minnesota Twins · vs RHP · 1 HR / 20 PA	10.9%	16.6%	16.7%	18.0%	16.6%	16.7%	20.2%	27.3%	17.7%	17.8%
10. Jonathan Aranda Tampa Bay Rays · vs RHP · 21 HR / 634 PA	14.7%	16.6%	15.7%	15.6%	16.6%	16.5%	15.8%	15.4%	15.5%	15.6%
11. Rafael Flores Jr. Pittsburgh Pirates · vs RHP · 9 HR / 135 PA	13.0%	16.4%	17.5%	16.5%	16.4%	16.4%	17.2%	16.8%	17.3%	17.4%
12. Brandon Lowe Pittsburgh Pirates · vs RHP · 31 HR / 616 PA	14.5%	16.3%	16.7%	16.8%	16.3%	16.3%	16.4%	17.3%	16.8%	16.9%
13. Zack Gelof slate Athletics · vs LHP · 18 HR / 393 PA	13.9%	16.1%	16.0%	16.2%	16.1%	16.1%	16.1%	16.0%	16.2%	16.3%
14. Carter Jensen Kansas City Royals · vs RHP · 24 HR / 567 PA	14.0%	16.1%	16.2%	15.6%	16.1%	16.0%	15.1%	15.7%	15.9%	16.0%
15. Francisco Alvarez New York Mets · vs RHP · 15 HR / 392 PA	12.9%	16.0%	15.6%	15.7%	16.0%	16.0%	15.7%	16.1%	16.1%	16.2%
16. Elly De La Cruz Cincinnati Reds · vs RHP · 28 HR / 593 PA	14.9%	15.9%	17.3%	16.3%	15.9%	15.9%	16.0%	16.5%	16.4%	16.5%
17. Yordan Alvarez Houston Astros · vs RHP · 39 HR / 646 PA	13.3%	15.7%	17.2%	16.6%	15.8%	15.8%	16.2%	17.6%	16.7%	16.8%
18. Munetaka Murakami Chicago White Sox · vs RHP · 31 HR / 508 PA	13.0%	15.5%	15.8%	16.0%	15.5%	15.6%	15.2%	15.5%	15.6%	15.7%
19. Sal Stewart Cincinnati Reds · vs RHP · 31 HR / 657 PA	14.2%	15.3%	16.0%	15.9%	15.3%	15.3%	15.4%	15.1%	15.5%	15.6%
20. Willson Contreras Boston Red Sox · vs RHP · 27 HR / 530 PA	12.4%	15.2%	14.8%	14.9%	15.2%	15.2%	15.2%	16.6%	15.2%	15.3%
21. Corey Seager Texas Rangers · vs LHP · 18 HR / 394 PA	12.3%	15.1%	14.7%	14.9%	15.1%	15.1%	14.8%	15.1%	15.2%	15.3%
22. Mickey Moniak	12.7%	14.9%	16.2%	15.3%	15.0%	14.9%	14.7%	15.5%	15.3%	15.4%

Hitter	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression	Neural net (torch)	X
37. Fernando Tatis Jr. San Diego Padres · vs RHP · 23 HR / 664 PA	13.5%	14.0%	13.5%	13.4%	13.9%	13.9%	14.1%	14.4%	14.3%	
38. Bryce Harper Philadelphia Phillies · vs RHP · 28 HR / 648 PA	12.7%	13.8%	14.2%	14.3%	13.8%	13.9%	13.8%	13.9%	14.1%	
39. Wilyer Abreu Boston Red Sox · vs RHP · 24 HR / 644 PA	12.4%	13.8%	14.0%	14.3%	13.8%	13.8%	13.6%	14.1%	14.0%	
40. Bobby Witt Jr. Kansas City Royals · vs RHP · 18 HR / 590 PA	12.3%	13.8%	13.2%	13.6%	13.8%	13.8%	13.4%	14.0%	13.6%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.